Avoid these five mistakes when training your AI

Meticulous teaching is the fundamental requirement for having an excellent performance consistently. However, there are a few mistakes to which traditional (and contemporary) data analytics or statistical methods are susceptible. Collective calling for such issues is often known as garbage in, garbage out.

1. Not having enough data to train

How much data does one need to train the AI system effectively? Well, it depends!

It's not the answer you would expect when you are at the pointy end of your machine learning stage. Nevertheless, it does depend on the complexity of your problem and the complexity of the algorithm you plan to use. Either way, the best way would be to use empirical investigation and arrive at an optimal number.

You may want to use standard sampling methods in the collection of required data, and may wish to use standard sample size calculators as used in standard statistical analysis tools. However, due to the nature of machine learning algorithms, the amount of data is often insufficient. You would most likely need more than what a standard sample size calculation formula tells you.

Having more data may not be a big problem as having less data. You have to make sure that there is enough data to reasonably capture the relationship that might exist within input parameters (features) and between input and output.

You may also use your domain expertise to reasonably assess how much data is enough to exhibit a full cycle of your business problem. It should cover all the possible seasonality and variations.

The model developed with the help of this data will only be as good as the data you have or provide for training, so make sure that it is adequately available. If you feel that the data is not enough, which may be a rare scenario in the current Big Data world, don't rush and wait until you get enough of it.

2. Not cleaning and validating the dataset

Too much data is of no use if it is of poor quality. It could mean one or more of the following three things.

Data has noise: There is too much conflicting and misleading information. Confounding variables or parameters are present, and the essential variables are missing. Cleaning this type of data needs additional data points because the current set is unusable and hence not enough.

It is dirty data: Several values are missing (though parameters are available), or the data has inconsistencies, errors, and a mix of numerical or categorical values in the same column. This type of data needs careful manual cleaning by subject matter experts and may often need re-validation. Depending on resource availability, you may find it easier to obtain additional data instead of cleaning dirty data.

Inadequate or sparse data: This is a scenario where very few data points have actual values, and a significant part of the dataset is full of nulls or zeroes.

The type of issues present within the dataset is often not clear from the dataset itself, which is why I always recommend exploratory analysis and visualisation to be applied at the outset. Doing this first not only gives you a level of confidence in data quality but also tells you if there is something amiss.

Based on the visual representation, an interesting question would be—do you see what you expect to see?

If the answer is 'no,' then your data may be of poor quality and needs cleaning.

If the answer is 'yes,' it might be useful in finding some preliminary insights. This validation of the dataset is essential to proceed, and you should never miss it.

3. Not having enough spread in data

Having a large amount of data is not always a good thing unless it can represent all the possible use cases or scenarios. If the data is missing variety, it can lead to problems in future—you increase the chances of losing on low-frequency high-risk scenarios.

For traditional predictive analysis, there is a point of low returns as you obtain more and more data for training. Your data science team can often spot this point empirically.

However, since machine learning is an inductive process, your base model can only cover what it has seen in the data. So, if you miss on edge cases, these will not be supported by your model. It merely means your AI will fail when that scenario occurs. This is the only and the most crucial reason why your training data should have enough spread to represent the real population.

4. Ignoring near-misses and overrides

During initial training, it is hard to identify near-misses and disregarded data points. However, in a continuous learning loop with feedback, it becomes highly essential to pay close attention to near-misses, and human or machine overrides.

When you deploy your AI system for the first time, it has an only base model that governs the performance of an AI. However, as system operation continues, the feedback loop feeds live data, and the system starts to adjust, either live or regularly.

If the model has missed to correctly predict or calculate any output just by a bit and thereby the decision has changed, it would be a near-miss. For example, in case of a loan approval system, if 88.5 per cent score means 'loan approved' and 88.6 per cent